

UNCONSTRAINED DYSFLUENCY MODELING FOR DYSFLUENT SPEECH TRANSCRIPTION AND DETECTION

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ABSTRACT

Dysfluent speech modeling requires time-accurate and silence-aware transcription at both the word-level and phonetic-level. However, current research in dysfluency modeling primarily focuses on either transcription or detection, and the performance of each aspect remains limited. In this work, we present an unconstrained dysfluency modeling (UDM) approach that addresses both transcription and detection in an automatic and hierarchical manner. UDM eliminates the need for extensive manual annotation by providing a comprehensive solution. Furthermore, we introduce a simulated dysfluent dataset called VCTK⁺⁺ to enhance the capabilities of UDM in phonetic transcription. Our experimental results demonstrate the effectiveness and robustness of our proposed methods in both transcription and detection tasks.

Index Terms: dysfluent speech, transcription, detection

1. INTRODUCTION

In the field of speech analysis, a clear and universally agreed-upon definition of dysfluent speech is yet to be established. Dysfluencies are commonly associated with speech disorders such as stuttering, aphasia [1], and dyslexia [2], characterized by disruptions in the normal flow of speech [3]. However, dysfluency also appears in normal conversational speech or spontaneous speech [4], where individuals may experience hesitations or interruptions while speaking. In this context, we refer to dysfluent speech as any form of speech characterized by abnormal patterns such as repetition, prolongation, and irregular pauses, as discussed in [5]. Within the domain of *dysfluent speech modeling*, there is also a lack of a unified paradigm. Research efforts in this area can be roughly categorized as *transcription* and *detection*.

Current state-of-the-art word transcription models [6–10] and phonetic transcription models [11] often struggle to accurately transcribe dysfluent speech. As a result, human annotations are still required and transcribers commonly resort to a two-step process. They first obtain automatic transcriptions and then manually annotate the missing portions, a laborious and time-consuming task that is comparable to manual annotation from scratch. WhisperX [12] recently extends Whisper [6] by generating timestamps for individual words. However, it still delivers limited performance for dysfluent speech. Moreover, time-accurate *phonetic* transcriptions might be a better representation to capture various dysfluency types. Another requirement is that phonetic transcriptions should be sensitive to silences as it might indicate a block or poor breath-speech coordination. [5] recently proposed a time-accurate and silence-aware neural forced

aligner, where a weighted finite-state transducer (WFST) is introduced for modeling dysfluency patterns such as repetition. However, this approach assumes that there is minimal deviation between the reference and “real” transcribed text. In real-life dysfluent speech, such as the example shown in Figure 1, this assumption may not hold true.

Research on dysfluency detection has traditionally been conducted independently of dysfluency transcription and has recently been dominated by end-to-end methods. These approaches typically focus on either utterance-level detection [13–16], or frame-level detection [17, 18]. However, these studies primarily address data-driven classification problems and do not explicitly incorporate dysfluency transcription into their detection methods. A unified framework that integrates dysfluency transcription and detection is essential to develop an efficient and robust dysfluency modeling system.

In this study, we propose an *unconstrained dysfluency modeling* (UDM) approach that integrates dysfluent speech transcription and pattern detection in an automatic manner with no human effort. Since real dysfluent speech is unconstrained and word transcription is unknown (as shown in the “Human Transcription” in Figure 1, which is largely different from reference text), we develop hierarchical transcription methods. Firstly, we introduce an unconstrained forced aligner with a dynamic alignment search module to generate text-independent alignments. Secondly, we propose a Text Refresher that leverages the alignment input to refine the state-of-the-art Whisper [6] transcription. For the detection component, we employ *2D-alignment* to automatically detect various phonetic and word-level dysfluency patterns, including repetition, missing, replacement, insertion, deletion, and irregular pauses. To further enhance performance, we curate a dysfluent dataset called VCTK⁺⁺ to boost the capacity of our unconstrained forced aligner (UFA). Experimental results demonstrate the effectiveness of our proposed framework in both dysfluent speech transcription and dysfluency pattern detection.

2. UNCONSTRAINED DYSFLUENCY TRANSCRIPTION

2.1. Unconstrained Forced Aligner (UFA)

Unconstrained forced aligner predicts alignment with weak text supervision. As shown in Fig. 2, a speech segment is passed into WavLM [19] encoder which generates latent representations. A conformer module [20] is followed to predict both alignment and boundary information. The alignment and boundary targets used in UFA are derived from the Montreal Forced Aligner (MFA) [21]. This integration with MFA allows UFA to operate with weak text supervision. During the inference stage, there is no need for text input, making the

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¹Audio sample for Fig.1:<https://shorturl.at/rTW26>

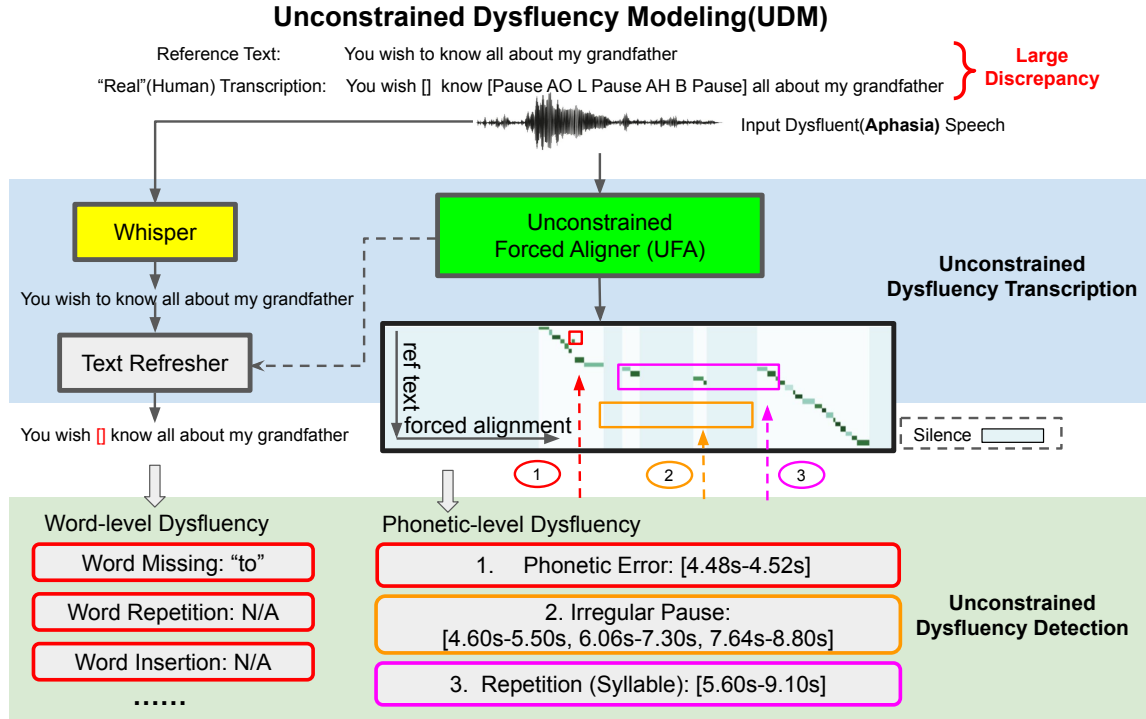


Fig. 1: Unconstrained Dysfluency Modeling (Transcription and Detection) for aphasia speech instance. Here is an example of aphasia speech. The reference text is "You wish to know all about my grandfather," while the human transcription or ground truth differs significantly from the reference. Whisper recognizes it as perfect speech, while UFA is able to capture most of the dysfluency patterns. An audio sample of this can be found here¹.

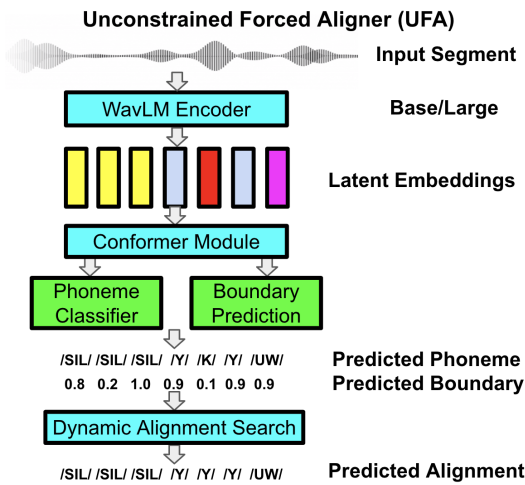


Fig. 2: UFA Module

alignment process unconstrained. The conformer module comprises of four conformer [20] encoder layers. The hidden size, number of attention heads, filter size, and dropout for each conformer layer are [1024, 4, 5, 0.1], [1024, 8, 3, 0.1], [1024, 8, 3, 0.1], [1024, 4, 3, 0.1] respectively. Two linear layers are simply applied as phoneme classifier and boundary predictor. For the phoneme classifier, UFA op-

timizes the softmax cross-entropy objective, while logistic regression is utilized for boundary prediction. Specifically, it predicts floating numbers between 0 (non-boundary) and 1 (boundary).

2.1.1. Dynamic Alignment Search

For the inference of dysfluent speech, such as aphasia speech, obtaining an accurate text transcription is often not achievable (as shown in the "Human Transcription" in Figure 1). Consequently, alignment should be decoded without text supervision. We propose a boundary-aware viterbi dynamic alignment search algorithm, which is the extension of Viterbi algorithm. Let us denote the phoneme logits as $\logits \in \mathbb{R}^{B,T,D}$, the boundary predictions as $boundaries \in \mathbb{R}^{B,T}$, and the bi-gram phoneme language model as $transition_probs \in \mathbb{R}^{D,D}$, where (B, T, D) represents the batch size, time steps, and phoneme dictionary size, respectively. The search algorithm is outlined in Algorithm 1.

It is worth noting that the transitions between consecutive phonemes near the boundary should be assigned lower importance to mitigate the risk of phoneme omissions. For instance, consider the correct alignment as SIL SIL SIL Y Y Y. In some cases, when the predicted probability for "Y" is low, there is a possibility that the prediction of "Y" might be overlooked due to the higher self-transition probability of SIL. Consequently, the final prediction could erroneously become SIL SIL SIL SIL SIL SIL. The bi-gram phoneme language model is derived by applying maximum likelihood estimation to the VCTK [22] forced alignment obtained from MFA [21].

Algorithm 1 Boundary-Aware Dynamic Alignment Search

```
1: procedure DECODE(logits, boundaries, transitional_probs)
2:    $B, T, D \leftarrow \text{shape of } \textit{logits}$ 
3:   Initialize trellis and backpointers
4:   for  $t$  in range(1,  $T$ ) do
5:     for  $d$  in range( $D$ ) do
6:        $\textit{trellis}[:, t, d], \textit{backpointers}[:, t, d] \leftarrow$ 
        $\text{MAX\_ARGMAX}(\textit{trellis}[:, t-1, :] + (1 - \textit{boundaries})[:,$ 
        $t] \times \textit{transitional\_probs}[d, :])$ 
7:     end for
8:   end for
9:   Derive best_path from trellis and backpointers
10:  return best_path
11: end procedure
```

2.2. ASR Transcription and Text-Refresher

State-of-the-art ASR models [6, 8, 23] are commonly trained using a robust language model constraint, ensuring a high level of accuracy in transcribing dysfluent or disordered speech, thereby generating nearly perfect transcriptions. However, to perform word-level dysfluency analysis, it is necessary to introduce imperfections. In this study, we propose *Text Refresher* to achieve this objective.

First, we obtain a perfect transcription using Whisper-large [6]. We then obtain its corresponding phoneme transcription using CMU dictionary [24]. Subsequently, in *Text Refresher*, we perform Dynamic Time Warping (DTW) between the phoneme transcription of the Whisper output and the output of the Unsupervised Forced Aligner (UFA). Our primary focus is on identifying *insertions* and *deletions*. If a word (represented as a phoneme sequence) in the Whisper output does not align with the correct word (phoneme sequence) in the UFA output, we remove that word. For example, in the case illustrated in Figure 1, the word "to" is deleted. On the other hand, if a word (phoneme sequence) in the UFA output does not align with any word (phoneme sequence) in the Whisper output, we insert that word. Our observations indicate that in Aphasia speech, most word-level imperfections that Whisper cannot transcribe are primarily from deletions or insertions.

2.3. Unconstrained Transcription Evaluation

2.3.1. Phonetic Transcription(Alignment) Evaluation

In order to evaluate how accurately the speech is transcribed at the frame level, we report **Micro F1 Score** and **Macro F1 Score** of phoneme transcription. Note that our F1 scores evaluate how many phonemes are correctly predicted. This is different from [25] which evaluates how many time steps are correctly predicted as phonetic boundaries. In order to evaluate the phoneme segmentation performance within our methods, we propose the duration-aware phoneme error rate (**dPER**). dPER extends Phoneme Error Rate (PER) by weighing each operation (substitution, insertion, deletion) with its duration. Denote $\hat{S}, \hat{I}, \hat{D}, \hat{C}$ as the weighted value of substitutions, insertions, deletions, and correct samples. Denote p_i and p_j as the current two phonemes we are comparing in the reference sequence and prediction sequence respectively. Denote $d(p_i)$ and $d(p_j)$ as their durations (number of repetitions). Whatever the error type is detected, we propose the following updating rule: $\hat{S} \rightarrow \hat{S} + d(p_i) + d(p_j)$, $\hat{I} \rightarrow \hat{I} + d(p_j)$, $\hat{D} \rightarrow \hat{D} + d(p_i)$, $\hat{C} \rightarrow \hat{C} + |d(p_i) - d(p_j)|$. The

ultimate formula is:

$$\text{dPER} = \frac{\hat{S} + \hat{D} + \hat{I}}{\hat{S} + \hat{D} + \hat{C}} \quad (1)$$

2.3.2. Imperfect Word Transcription Evaluation

In contrast to conventional ASR tasks, evaluating the performance in word-level dysfluency analysis requires the utilization of imperfect word targets. In this study, we employ manual word annotation of Aphasia speech as the target reference and report the *imperfect Word Error Rate* (**iWER**).

3. UNCONSTRAINED DYSFLUENCY DETECTION

We develop rule-based methods for detecting time-accurate phonetic-level dysfluencies, including *Phonetic Errors* (*Missing*, *Deletion*, *Replacement*), *Repetition*, and *Irregular Pause*. Additionally, our methods also cover word-level dysfluencies, including *Missing*, *Insertion*, *Replacement*, and *Repetition*.

3.1. Phonetic-Level Dysfluency Detection

Phonetic-level dysfluency detection is empirically performed by comparing the predicted forced alignment with the reference ground truth text. The initial and crucial step in this process involves accurately **aligning the forced alignment with the reference text**. This alignment necessitates non-monotonicity whenever a phonetic dysfluency, such as repetition or insertion, occurs. Traditional aligners, such as Dynamic Time Warping (DTW) and MFA [21], cannot handle non-monotonic cases and are inadequate for this purpose.

To address this limitation, we extract the phoneme center embeddings from the phoneme classifier in the Unconstrained Forced Aligner (UFA) (Fig. 2). By obtaining the phoneme embedding sequences for both the reference text and the forced alignment, we compute the dot product between these sequences. As a result, we generate a 2D similarity matrix that serves as the alignment representation. In the forced alignment, each phoneme may align with multiple occurrences of the same phoneme in the reference text, particularly when the reference text contains repeated phonemes. For instance, in the phrase "Please call Stella" represented as "P L I Y Z K A O L S T E H L A H," each occurrence of "L" in the forced alignment aligns with all three "L" phonemes in the reference text. To ensure that only one phoneme in the reference aligns with the current phoneme in the forced alignment, we employ Viterbi search on the 2D similarity matrix. This process yields the final 2D alignment, denoted as **alignment-2d**, which is primarily monotonic. As illustrated in Figure 1 and Figure 3, the alignment-2d is visualized through green plots, highlighting the relationship between the forced alignment and the reference text.

In addition to the alignment-2d, we also require a ground truth alignment-2d, which represents the expected alignment between the forced alignment of nearly perfect speech and the reference text. This ground truth alignment is strictly monotonic. To obtain it, we apply Dynamic Time Warping (DTW) between the forced alignment and the reference text, resulting in the alignment represented by the red plots in Figure 3. We denote this as **alignment-2D-DTW**.

Finally, the detection of phonetic dysfluency becomes straightforward with the availability of the alignment-2D and alignment-2D-DTW. As illustrated in Figure 3, in the case of normal speech, these two alignments align perfectly with each other. However, if there is a significant decrease in the alignment-2D-DTW while lacking any intersection in the corresponding row, it indicates a **missing**

phoneme, as depicted in Fig 3 (b). If a row in alignment-2D-DTW encounters multiple columns in alignment-2D, and there are repeated phonemes present, it indicates a **repetition**. This is depicted in Figure 3 (d). Conversely, if a row in alignment-2D-DTW already aligns with alignment-2D and simultaneously aligns with the surrounding column in alignment-2D, it signifies an **insertion**. This is illustrated in Figure 3 (c). If a row in alignment-2D-DTW does not overlap with any horizontal regions in alignment-2D, but only overlaps with a single vertical block in alignment-2D, it is recognized as a **replacement**. This is depicted in Figure 3 (e). Lastly, any pauses occurring within a complete sentence are identified as **irregular pauses**, as shown in Figure 3 (f). It should be noted that within this rule-based detection framework, the precise timing of all five types of dysfluencies can be accurately identified with a resolution of 20ms.

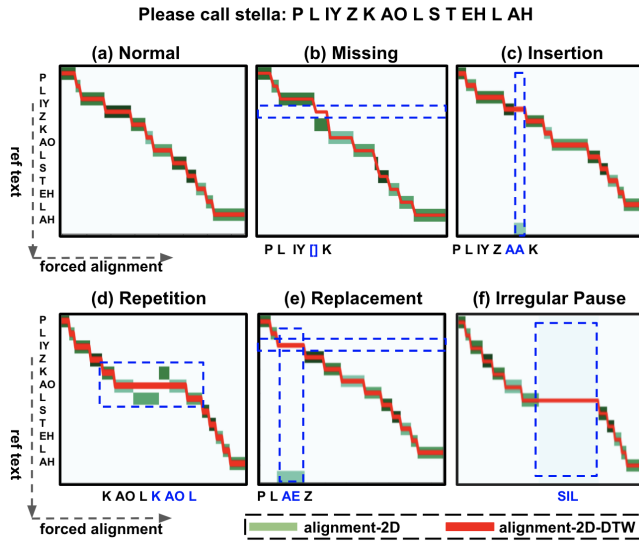


Fig. 3: Phonetic-Level Dysfluency Detection. Audio samples can be found here.²

3.2. Word-level Dysfluency Detection

We address *missing*, *insertion*, *replacement*, and *repetition* as part of our word-level dysfluency detection. To detect word-level dysfluency, we follow a similar methodology as phonetic-level dysfluency detection, which involves obtaining **alignment-2D** and **alignment-2D-DTW**. However, in the case of word-level dysfluency detection, we do not utilize word embeddings. Instead, we employ perfect matching between the words in the reference and predicted texts, without the need for embedding dot product calculations. Duration, including silence, is not taken into account in this particular analysis since it is already incorporated in the phonetic component.

3.3. Dysfluency Evaluation

We conduct dysfluency evaluation on segments of Aphasia speech. In each Aphasia speech segment, manual annotations are made for all types of dysfluencies, including their accurate timing. For the evaluation of phonetic-level dysfluency, we report the **F1 score (Micro and Macro)** for dysfluency type identification. Additionally, we measure the accuracy of dysfluency detection in terms of time alignment. To achieve this, we calculate the Intersection over Union (IoU) between

our predicted time boundaries and the ground truth boundaries. If the IoU is greater than 0.5, the dysfluency is identified as detected. We also report the F1 score for this matching evaluation, referred to as the **matching score**. For the evaluation of word-level dysfluency, we simply report the F1 score (Micro and Macro) without considering the timing aspects.

4. EXPERIMENTS

4.1. Datasets and Pre-processing

VCTK [22] It is a multi-speaker accented corpus containing 44 hours of fluent speech. Following [26–28], we randomly select 90% of speakers as training set and the remaining as dev set. VCTK is used to train *Unconstrained Forced Aligner*.

VCTK⁺⁺ For each waveform in VCTK and its forced alignment (from MFA [21]), we applied simulations regarding the following stutter types. **(i) Repetitions:** Phonemes are randomly sampled within the waveform, appended by a variable-length sample of silence, and inserted into the original sound file. The silence sample is set to vary between 200ms and 500ms in multiples of 20 to match the framerate of the phoneme alignments. **(ii) Prolongations:** Phonemes are randomly selected, excluding phonemes that cannot be reasonably prolonged, such as hard consonants or silence tokens. The sound sample containing the phoneme is then stretched by a random factor anywhere from 5x to 10x using Waveform Similarity Overlap-Add (WSOLA) [29]. The original phoneme is then replaced by the stretched variant in the waveform. **(iii) Blocks:** Phonemes are selected from a list of commonly blocked sounds, such as consonants or combinations of hard phonemes. With each simulation, we maintain the phoneme alignments such that the phoneme timestamps line up with the individual stutters, generating new alignments that act as ground truth for inference. See supplemental material for details. Here is an example of our augmented data. <https://shorturl.at/aeGKR>

Buckeye [4] It contains over 40 hours of recordings from 40 native speakers of American English. The corpus contains quite a few portions of dysfluent speech with time-accurate annotation. We follow [25] to make the train/val/test split. The Buckeye corpus is utilized for training the *Unconstrained Forced Aligner* and for *Phonetic Transcription Evaluation*.

Aphasia Speech From our clinical collaborators, our dysfluent data comprises ten participants diagnosed with Primary Progressive Aphasia (PPA), which exhibits various manifestations of dysfluencies. The dataset consists of audio recordings capturing interactions between patients and speech-language pathologists (SLPs). Our primary focus lies in the audio input of patients reading the Grandfather passage, resulting in approximately 20 minutes of speech data. The Aphasia speech dataset is employed for the evaluation of *Imperfect Word Transcription and Dysfluency Detection*.

Phonetic Dictionary We remove stress-aware phoneme labels (e.g. AE0, AE1→AE). The phoneme dictionary adopted in this paper contains 39 monophones from CMU phoneme dictionary [24] along with one additional silence label. For Buckeye corpus, we manually translate the out-of-dictionary phonemes into CMU monophones. Here is the translation paradigm: AEN→AE N, EYN→EY N, IYN→IY N, TQ→T, IHN→IH N, OWN→OW N, NX→N, EHN→EH N, DX→T,

²Audio sample for Fig.3:<https://rb.gy/fmiv7>

Method	WavLM Size	Training Data	Micro F1 (% , $\uparrow$)	Macro F1 (% , $\uparrow$)	dPER (% , $\downarrow$)	Micro F1 (% , $\uparrow$)	Macro F1 (% , $\uparrow$)	dPER (% , $\downarrow$)
			<i>Buckeye Test Set</i>			<i>VCTK++ Test Set</i>		
WavLM-CTC-VAD	Large	None	50.1	47.3	86.9	48.8	45.7	88.0
WavLM-CTC-MFA	Large	None	49.8	28.7	53.9	47.6	26.0	54.2
UFA	Base	VCTK	68.9	55.6	53.3	78.8	59.5	53.4
UFA	Base	VCTK+Buckeye	65.9	51.6	63.6	75.2	56.0	60.0
UFA	Large	VCTK+Buckeye	70.3	55.0	46.2	80.7	66.4	45.8
UFA	Large	VCTK	71.3	60.0	46.0	81.7	72.0	44.0
– Boundary-aware	Large	VCTK	68.9	52.0	49.9	78.4	62.9	47.8
UFA	Large	VCTK ⁺⁺	73.5	64.0	41.0	93.6	90.8	38.0
– Boundary-aware	Large	VCTK ⁺⁺	71.0	63.7	44.3	91.1	90.0	42.1

Table 1: Phonetic transcription evaluation on two dysfluent corpora: Buckeye and VCTK++ test set.

Method	iWER (% , $\downarrow$)
Whisper-Large [6]	11.3
+Text Refresher	9.7
+VCTK ⁺⁺	9.2

Table 2: Word Transcription Evaluation on Aphasia Speech

EN→AH N, OYN→OY N, EM→EH M, ENG→EH NG, EL→EH L, AAN→AA N, AHN→AH N, AWN→AW N.

Audio Segmentation For VCTK, we train on the entire utterance without segmentation. For Buckeye data, we follow [25] to segment the long utterance by the ground truth transcription. We make sure that the beginning and ending silence length would be no longer than 3s, resulting in the length of all segments ranging from 2s to 17s. Different from [25], we keep all silence labels but still remove the untranscriptable labels such as 'LAUGH', 'IVER', etc. For patient speech, we apply the online Silero VAD [30] with a default threshold of 0.5 to make the segments. We keep all of the silences and this results in the length of all segments ranging from 2s to 15s. All audio samples have a sampling rate of 16K Hz.

4.2. Phonetic Transcription Experiments

We train our unconstrained forced aligner (UFA) using three types of data: VCTK only, VCTK+Buckeye, and VCTK++. Additionally, we conduct an ablation study to examine the impact of the boundary-aware constraint in the dynamic search algorithm. This is achieved by removing the constraint from the search algorithm. Across all experiments, we utilize the same configuration settings, employing the Adam optimizer with an initial learning rate of 1e-3, which is decayed by 0.9 at each step. Each model converges after approximately 10 epochs, as determined by achieving a 90% phoneme classification accuracy on the development set.

Furthermore, we investigate two alternative forced aligners for comparison purposes: WavLM-CTC-VAD and WavLM-CTC-MFA. In WavLM-CTC-VAD, we combine the CTC phoneme alignment [31] obtained from WavLM-CTC [11] with Voice Activity Detection (VAD) segmentation. By assigning blank tokens and incorporating silence segments identified using online Silero VAD [30], we obtain a silence-aware transcription. The VAD threshold is set to the default value of 0.5, and the minimum and maximum speech durations are defined as 250ms and infinity, respectively. In WavLM-CTC-MFA, we employ the Montreal Forced Aligner (MFA) [21] to derive silence-aware phoneme alignment. We utilize WavLM-CTC [11] to

generate the initial phoneme transcription, and we leverage a pre-trained English ARPA acoustic model. A pronunciation dictionary maps phonemes (as word-level items) to phonemes (as phonemic pronunciation breakdowns). The default beam size of 10 is applied for MFA. In the phoneme-to-phoneme dictionary, the parameters for each phoneme mapping include a pronunciation probability of 0.99, a silence probability of 0.05, and final silence and non-silence correction terms of 1.0. For both methods, no additional training data is needed. Phonetic transcription results are shown in Table. 1.

4.3. Imperfect Word Transcription Experiments

We utilize Whisper-large [6] as our ASR transcriber. We begin by presenting the results obtained directly from Whisper-large. Subsequently, we employ Text Refresher to refine the Whisper transcription and report the updated results. By default, Text Refresher incorporates the UFA-WavLM-Large-VCTK alignment. Additionally, for ablation purposes, we consider the UFA-WavLM-Large-VCTK⁺⁺ alignment as input, which demonstrated superior performance as indicated in Table 1. The comprehensive results are presented in Table 2.

4.4. Dysfluency Detection

Methods	F1 (% , $\uparrow$)	MS (% , $\uparrow$)	Human F1 (% , $\uparrow$)	Human MS (% , $\uparrow$)
UFA-VCTK	62.4	55.2	90.4	85.6
UFA-VCTK ⁺⁺	64.5	60.2	90.6	86.0

Table 3: Phonetic Dysfluency evaluation on Aphasia speech.

The preliminary experiments presented in Table 1 indicate that both WavLM-CTC-VAD and WavLM-CTC-MFA do not exhibit significant improvements in phonetic transcription performance. Furthermore, the joint training of the VCTK and Buckeye corpora does not enhance the overall performance. Hence, we restrict our evaluation to two variants of the Unconstrained Forced Aligner (UFA): UFA-WavLM-Large-VCTK and UFA-WavLM-Large-VCTK⁺⁺. To assess the efficacy of our rule-based detection algorithm, we also perform manual detection using the predicted alignment from UFA and human-created targets. The results are presented in Table 3 and Table 4. MS is *Matching Score*, as stated in Sec. 3.3.

Methods	F1 (% , $\uparrow$)	Human F1 (% , $\uparrow$)
Whisper-Large [6]	64.0	86.4
+Text Refresher(VCTK)	66.8	88.0
+Text Refresher(VCTK ⁺⁺)	68.4	89.1

Table 4: Word Dysfluency evaluation on Aphasia speech.

4.5. Results and Discussion

4.5.1. Transcription Analysis

We begin by examining the phonetic transcription results, as presented in Table 1. Both WavLM-CTC-VAD and WavLM-CTC-MFA demonstrate commendable zero-shot silence-aware phonetic transcription capabilities. However, their performance remains limited and is even inferior to the UFA trained with the WavLM base model. Interestingly, incorporating the Buckeye data during training does not yield any performance improvement. We hypothesize that the presence of noise in the Buckeye corpus, being a dysfluent dataset itself, hinders performance. Additionally, including the LibriSpeech dataset in VCTK training does not lead to performance enhancement. This suggests that UFA has already reached a certain limit in terms of data scalability. Consequently, the subsequent ablations and dysfluency detection experiments are conducted solely using UFA-WavLM-Large-VCTK. During our ablation study, we consistently observed performance improvements by incorporating boundary prediction information in the dynamic alignment search, as described in Section 2.1.1. Moreover, our experiments on VCTK⁺⁺ consistently demonstrated enhanced performance compared to the original VCTK dataset, highlighting the robustness introduced by VCTK⁺⁺. In terms of word transcription results, as shown in Table 2, we found that Whisper-Large exhibited the lowest performance due to its overpowering language modeling. However, with the introduction of Text Refresher and the incorporation of VCTK⁺⁺, we observed an improvement in the imperfect Word Error Rate (iWER), further boosting the overall performance.

4.5.2. Dysfluency Analysis

Since there is no previous work on hierarchical (word/phoneme) and fine-grained (time-accurate) dysfluency detection models like ours, we conducted ablation experiments to compare our proposed rule-based detection methods against ourselves. The results, as shown in Table 3 and Table 4, indicate impressive performance in terms of F1 scores and matching scores (MS), demonstrating the ability of our methods to accurately capture most dysfluencies. However, it is important to note that our methods still fall short of human detection performance, highlighting their inherent limitations.

5. CONCLUSION AND LIMITATIONS

We propose an unconstrained dysfluency modeling approach that combines transcription and detection, which has been proven effective in both tasks. However, there are several limitations that should be addressed in future research. First, our detection experiments primarily focus on aphasia speech, which limits the generalizability. Future work should explore diverse and open-domain dysfluent datasets, which may lack manual annotations. Second, our approach relies on phoneme-level forced alignment as the key representation for detection. However, it is worth investigating alternative speech units, such as self-supervised discrete units [32] and articulatory units [33–36], to improve alignment modeling. Third, while our detection method

performs well on "easy" speech samples, it struggles with more challenging and abnormal speech patterns. Even in the current setting, the detection scores are still below human performance. Lastly, it is worth exploring the application of LLM-guided speech models [37, 38] to advance dysfluency modeling in a prompt manner, which remains an open problem. Hopefully our proposed approach can serve as a promising paradigm for the field of dysfluency modeling.

6. ACKNOWLEDGEMENT

Thanks for support from UC Noyce Initiative, Society of Hellman Fellows, NIH/NIDCD and the Schwab Innovation fund.

7. REFERENCES

- [1] Marian C Brady, Helen Kelly, Jon Godwin, Pam Enderby, and Pauline Campbell, "Speech and language therapy for aphasia following stroke," *Cochrane database of systematic reviews*, no. 6, 2016.
- [2] Margaret J Snowling and Joy Stackhouse, *Dyslexia, speech and language: a practitioner's handbook*, John Wiley & Sons, 2013.
- [3] Juraj Pálffy and Jiří Pospíchal, "Pattern search in dysfluent speech," in *2012 IEEE International Workshop on Machine Learning for Signal Processing*. IEEE, 2012, pp. 1–6.
- [4] Mark A Pitt, Keith Johnson, Elizabeth Hume, Scott Kiesling, and William Raymond, "The buckeye corpus of conversational speech: Labeling conventions and a test of transcriber reliability," *Speech Communication*, vol. 45, no. 1, pp. 89–95, 2005.
- [5] Theodoros Kouzelis, Georgios Paraskevopoulos, Athanasios Katsamanis, and Vassilis Katsouros, "Weakly-supervised forced alignment of disfluent speech using phoneme-level modeling," *Interspeech*, 2023.
- [6] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever, "Robust speech recognition via large-scale weak supervision," in *International Conference on Machine Learning*. PMLR, 2023, pp. 28492–28518.
- [7] Jiachen Lian, Alexei Baevski, Wei-Ning Hsu, and Michael Auli, "Av-data2vec: Self-supervised learning of audio-visual speech representations with contextualized target representations," *arXiv preprint arXiv:2302.06419*, 2023.
- [8] Yu Zhang, Wei Han, James Qin, Yongqiang Wang, Ankur Bapna, Zhehuai Chen, Nanxin Chen, Bo Li, Vera Axelrod, Gary Wang, et al., "Google usm: Scaling automatic speech recognition beyond 100 languages," *arXiv preprint arXiv:2303.01037*, 2023.
- [9] Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaocheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, et al., "Scaling speech technology to 1,000+ languages," *arXiv preprint arXiv:2305.13516*, 2023.
- [10] Armen Aghajanyan, Lili Yu, Alexis Conneau, Wei-Ning Hsu, Karen Hambardzumyan, Susan Zhang, Stephen Roller, Naman Goyal, Omer Levy, and Luke Zettlemoyer, "Scaling laws for generative mixed-modal language models," *International Conference on Machine Learning*, 2023.
- [11] "Wavlm-ctc-huggingface," <https://huggingface.co/microsoft/wavlm-large>.

- [12] Max Bain, Jaesung Huh, Tengda Han, and Andrew Zisserman, "Whisperx: Time-accurate speech transcription of long-form audio," *Interspeech*, 2023.
- [13] Tedd Kourkounakis, Amirhossein Hajavi, and Ali Etemad, "Fluentnet: End-to-end detection of stuttered speech disfluencies with deep learning," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 2986–2999, 2021.
- [14] Sadeen Alharbi, Anthony JH Simons, Shelagh Brumfitt, and Phil D Green, "Automatic recognition of children's read speech for stuttering application," in *6th. Workshop on Child Computer Interaction (WOCCI 2017)*, eds. K. Evanini, M. Najafian, S. Safavi and K. Berkling. International Speech Communication Association (ISCA), 2017, pp. 1–6.
- [15] Sadeen Alharbi, Madina Hasan, Anthony JH Simons, Shelagh Brumfitt, and Phil Green, "Sequence labeling to detect stuttering events in read speech," *Computer Speech & Language*, vol. 62, pp. 101052, 2020.
- [16] Melanie Jouaiti and Kerstin Dautenhahn, "Dysfluency classification in stuttered speech using deep learning for real-time applications," in *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 6482–6486.
- [17] John Harvill, Mark Hasegawa-Johnson, and Changdong Yoo, "Frame-level stutter detection," in *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, 2022, vol. 2022, pp. 2843–2847.
- [18] Olabanji Shonibare, Xiaosu Tong, and Venkatesh Ravichandran, "Enhancing asr for stuttered speech with limited data using detect and pass," *arXiv preprint arXiv:2202.05396*, 2022.
- [19] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, et al., "Wavlm: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [20] Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, and Ruoming Pang, "Conformer: Convolution-augmented Transformer for Speech Recognition," in *Proc. Interspeech 2020*, 2020, pp. 5036–5040.
- [21] Michael McAuliffe, Michaela Socolof, Sarah Mihuc, Michael Wagner, and Morgan Sonderegger, "Montreal forced aligner: Trainable text-speech alignment using kaldi," in *Interspeech*, 2017, vol. 2017, pp. 498–502.
- [22] Junichi Yamagishi, Christophe Veaux, Kirsten MacDonald, et al., "Cstr vctk corpus: English multi-speaker corpus for cstr voice cloning toolkit (version 0.92)," *University of Edinburgh. The Centre for Speech Technology Research (CSTR)*, 2019.
- [23] Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, et al., "Scaling speech technology to 1,000+ languages," *arXiv preprint arXiv:2305.13516*, 2023.
- [24] "Cmu phoneme dictionary," <http://www.speech.cs.cmu.edu/cgi-bin/cmudict>.
- [25] Luke Strgar and David Harwath, "Phoneme segmentation using self-supervised speech models," in *2022 IEEE Spoken Language Technology Workshop (SLT)*, 2023, pp. 1067–1073.
- [26] Jiachen Lian, Chunlei Zhang, and Dong Yu, "Robust disentangled variational speech representation learning for zero-shot voice conversion," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 6572–6576.
- [27] Jiachen Lian, Chunlei Zhang, Gopala Krishna Anumanchipalli, and Dong Yu, "Towards Improved Zero-shot Voice Conversion with Conditional DSVAE," in *Proc. Interspeech 2022*, 2022, pp. 2598–2602.
- [28] Jiachen Lian, Chunlei Zhang, Gopala K. Anumanchipalli, and Dong Yu, "Unsupervised tts acoustic modeling for tts with conditional disentangled sequential vae," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2548–2557, 2023.
- [29] W. Verhelst and M. Roelands, "An overlap-add technique based on waveform similarity (wsola) for high quality time-scale modification of speech," in *1993 IEEE International Conference on Acoustics, Speech, and Signal Processing*, 1993, vol. 2, pp. 554–557 vol.2.
- [30] Silero Team, "Silero vad: pre-trained enterprise-grade voice activity detector (vad), number detector and language classifier," <https://github.com/snakers4/silero-vad>, 2021.
- [31] Ludwig Kürzinger, Dominik Winkelbauer, Lujun Li, Tobias Watzel, and Gerhard Rigoll, "Ctc-segmentation of large corpora for german end-to-end speech recognition," in *Speech and Computer: 22nd International Conference, SPECOM 2020, St. Petersburg, Russia, October 7–9, 2020, Proceedings 22*. Springer, 2020, pp. 267–278.
- [32] Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed, "Hubert: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451–3460, 2021.
- [33] Jiachen Lian, Alan W Black, Louis Goldstein, and Gopala Krishna Anumanchipalli, "Deep Neural Convolutional Matrix Factorization for Articulatory Representation Decomposition," in *Proc. Interspeech 2022*, 2022, pp. 4686–4690.
- [34] Jiachen Lian, Alan W Black, Yijing Lu, Louis Goldstein, Shinji Watanabe, and Gopala K Anumanchipalli, "Articulatory representation learning via joint factor analysis and neural matrix factorization," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [35] Peter Wu, Li-Wei Chen, Cheol Jun Cho, Shinji Watanabe, Louis Goldstein, Alan W Black, and Gopala K. Anumanchipalli, "Speaker-independent acoustic-to-articulatory speech inversion," in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [36] Peter Wu, Tingle Li, Yijing Lu, Yubin Zhang, Jiachen Lian, Alan W Black, Louis Goldstein, Shinji Watanabe, and Gopala K. Anumanchipalli, "Deep Speech Synthesis from MRI-Based Articulatory Representations," in *Proc. INTERSPEECH 2023*, 2023, pp. 5132–5136.
- [37] Yuan Gong, Hongyin Luo, Alexander H Liu, Leonid Karlinsky, and James Glass, "Listen, think, and understand," *arXiv preprint arXiv:2305.10790*, 2023.

- [38] Paul K Rubenstein, Chulayuth Asawaroengchai, Duc Dung Nguyen, Ankur Bapna, Zalán Borsos, Félix de Chaumont Quitry, Peter Chen, Dalia El Badawy, Wei Han, Eugene Kharitonov, et al., “Audiopalm: A large language model that can speak and listen,” *arXiv preprint arXiv:2306.12925*, 2023.